

Ethan Ball (508) 446-2067 • ethanball29@gmail.com • Portfolio: starchy29.github.io/Portfolio

EDUCATION

Rochester Institute of Technology

Master of Science, Game Design and Development

Bachelor of Science, Game Design and Development, Summa Cum Laude

Rochester, NY

May 2024

May 2024

SKILLS

Programming Languages: C++, C#, HLSL, Java, Javascript, Python

Software: DirectX 11, 3D Math, Github, Perforce, HTML, CSS

Game Engines: Unity, Monogame, Playstation 4 Development Kit, Unreal Engine 4

GAME DEVELOPMENT EXPERIENCE

Changeling VR

Game Developer

Rochester, NY

Summer 2023

- Utilized Unreal Engine 4 blueprints to implement a diegetic UI and enhance player interaction
- Resolved technical problems for level designers to streamline content creation

Rockstar Games

AI Programmer Intern

Andover, MA

Summer 2022

- Developed a novel input system for potential game mechanics and debugging tools
- Investigated Rockstar's proprietary game engine to ideate possible features
- Gained proficiency in industry-standard C++ practices via mentorship and codebase observation

PROJECTS

Elbran Game Engine (Personal Project)

November 2024 - Present

- Architected core game engine systems for developing 2D games in C++
- Abstracted the platform layer to simplify porting and used the Win32 API to support Windows
- Utilized DirectX 11 and HLSL to implement a rendering pipeline with advanced graphical features including post-processing, particles, lighting, and text rendering from TrueType font files
- Refactored systems into a data-oriented approach to double the real-time frame rate
- Created a custom build process using batch files to script command line calls

Cauldron Craft (Personal Project)

December 2023 - November 2024

- Designed and developed a 2D turn-based tactics game in Unity using C#
- Implemented an artificial intelligence enemy system for automated computer opponents

PS4 Space Game (Academic Project)

December 2023

- Developed a simple space shooter game on a Playstation 4 development kit using C++
- Leveraged graphics programming knowledge to set up a rendering pipeline for game objects

ADDITIONAL EXPERIENCE

Volante Farms

Produce Handler, Cashier, Field Hand

Needham, MA

December 2024 - Present

- Collaborated with teammates to ensure a consistent quality and quantity of produce in stock
- Assisted customers in all aspects of their experience from answering questions to checking out
- Worked alongside the field crew to cultivate and harvest homegrown produce

Rochester Institute of Technology

Teaching Assistant

Rochester, NY

2022 - 2024

- Mentored students in C# programming and evaluated class assignments

Arbor Biotechnologies

Operations Coordinator Intern

Cambridge, MA

Summer 2021

- Supported scientific research operations and inventory management